

DEVILS ISLAND, WI, USA

WMO: 994190

Lat: 47.0795N

Lon: 90.7283W

Elev: 192

StdP: 99.04

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-18.6	-25.2	0.4	-20.0	-22.8	0.5	-17.4	17.9	-7.1	16.3	-8.8	8.8	270	0.753

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.9	25.9	20.6	24.3	19.6	23.0	18.8	22.0	24.1	20.8	22.9	19.9	22.1	4.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	16.3	23.2	20.2	15.2	22.3	19.1	14.2	21.1	65.4	24.2	61.4	22.9	57.8	22.1	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.3	13.4	-23.7	30.5	4.5	1.9	-27.0	31.8	-29.6	32.9	-32.2	34.0	-35.5	35.3		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-7.9	-7.3	-1.7	3.4	8.0	13.0	18.1	18.9	15.0	8.6	1.6	-3.8
	DBStd	10.14	6.18	5.90	5.53	3.54	3.54	3.53	3.52	2.72	3.71	3.86	4.91	4.82
	HDD10.0	2453	553	486	364	198	83	12	0	0	4	72	252	429
	HDD18.3	4758	812	719	621	447	319	163	47	27	113	302	501	687
	CDD10.0	836	0	0	2	2	22	102	252	274	152	29	1	0
	CDD18.3	99	0	0	0	0	0	3	41	43	11	1	0	0
	CDH23.3	262	0	0	0	0	3	16	126	82	33	2	0	0
	CDH26.7	35	0	0	0	0	1	3	18	9	4	0	0	0
Wind	WSAvg	6.3	7.3	7.5	6.6	6.7	5.8	4.7	4.7	5.0	5.5	6.8	7.2	7.2
Precipitation	PrecAvg	733	32	24	32	55	73	95	88	77	83	88	57	40
	PrecMax	955	67	57	80	205	127	205	206	152	182	219	113	88
	PrecMin	557	9	6	10	12	21	12	37	26	22	29	18	6
	PrecStd	106	15	13	17	40	27	42	43	30	37	48	27	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	6.9	16.5	16.1	22.6	25.6	29.0	28.0	26.9	22.0	13.0	6.1
		MCWB	3.7	3.3	13.0	9.6	15.2	19.5	20.9	21.7	20.5	16.7	9.6	4.5
	2%	DB	3.2	2.8	9.7	12.5	19.0	22.5	26.2	25.4	23.8	18.6	10.5	4.2
		MCWB	2.0	0.4	7.1	7.0	13.0	16.6	20.8	20.9	19.5	14.0	7.4	3.1
	5%	DB	1.7	1.1	6.2	10.2	16.0	20.4	24.6	23.9	21.6	15.8	9.1	3.1
		MCWB	0.8	-0.3	3.5	6.4	10.9	16.0	19.7	19.7	18.4	12.3	6.8	2.2
	10%	DB	0.2	0.1	4.1	7.8	13.7	18.6	23.0	22.7	20.1	13.5	7.9	2.3
		MCWB	-0.9	-1.0	2.2	4.9	9.6	15.2	18.9	19.1	17.2	10.8	6.0	1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	3.2	14.1	11.4	17.9	20.7	23.8	23.6	21.9	17.5	11.0	5.2
		MCDB	6.0	5.3	16.9	15.1	20.4	23.7	26.4	25.8	24.2	21.3	11.4	5.9
	2%	WB	2.6	0.8	6.8	8.4	14.4	18.4	22.1	22.0	20.3	14.4	8.5	3.6
		MCDB	3.6	1.7	9.1	11.6	17.6	20.0	24.3	24.0	22.7	17.3	9.7	4.0
	5%	WB	1.2	0.0	3.9	6.7	12.3	16.7	20.8	21.0	18.9	12.5	7.2	2.5
		MCDB	2.0	0.8	5.3	9.5	15.9	19.0	22.8	23.0	20.9	14.6	8.6	3.1
	10%	WB	-0.2	-0.9	2.6	5.3	10.6	15.5	19.7	20.1	17.6	11.1	6.0	1.6
		MCDB	0.8	0.0	3.8	7.7	13.5	17.8	22.1	22.0	19.4	12.8	7.6	2.2
Mean Daily Temperature Range	5% DB	MDBR	5.4	5.5	5.2	5.3	6.7	7.3	6.2	4.9	4.7	4.0	4.3	4.4
		MCDBR	5.7	5.8	7.6	10.1	11.7	10.7	8.4	6.8	7.4	7.8	5.3	4.7
		MCWBR	5.4	4.6	5.3	6.1	6.7	6.3	4.8	3.9	4.4	5.2	4.5	4.3
	5% WB	MCDBR	6.0	5.1	7.0	9.2	11.0	9.7	7.4	6.2	6.7	7.0	4.9	4.1
		MCWBR	5.7	4.5	5.5	6.1	7.9	7.1	5.1	4.4	4.7	5.0	4.5	4.0
Clear-Sky Solar Irradiance	taub	0.260	0.279	0.306	0.346	0.371	0.380	0.387	0.381	0.364	0.318	0.295	0.272	
	taud	2.506	2.426	2.401	2.373	2.403	2.418	2.440	2.466	2.482	2.567	2.540	2.515	
	Ebn at Noon	856	901	921	909	894	884	873	864	849	846	801	795	
	Edh at Noon	62	82	99	112	113	113	109	102	92	72	61	55	
All-Sky Solar Radiation	RadAvg	1.28	2.27	3.50	4.59	5.37	5.79	6.09	5.12	3.82	2.31	1.30	0.95	
	RadStd	0.14	0.23	0.32	0.48	0.39	0.40	0.24	0.34	0.23	0.27	0.17	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page